

Topology Lecture Notes: Hausdorff Spaces

Lecture by Zamir Hussain

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Introduction: Why We Need ‘Nice’ Spaces

In mathematics, we often add conditions to our definitions to ensure that the objects we study are “well-behaved” and match our intuition. In topology, one of the most important of these conditions is the **Hausdorff property**.

Intuition

The Hausdorff property is like having “enough room” in your space to separate any two distinct points with non-overlapping neighborhoods. Think of it as the topological version of being able to “draw a bubble” around each point without the bubbles touching.

To see why we need it, let’s look at a “pathological” or “weird” space that lacks this property. This example will show us the kinds of bizarre behavior we want to avoid.

1 A ‘Weird’ Space: Motivation for Separation

A Non-Hausdorff Space

Consider a space Z with just three points: $Z = \{1, 2, 3\}$. Let’s define a topology τ on Z with the following open sets:

$$\tau = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}\}$$

This is a valid topology. Let’s see what goes wrong with it.

Problem 1: Points Are Not Necessarily Closed

In familiar spaces like $\mathbb{R}$, a single point is always a closed set. Is that true here? Let’s check the set $\{1\}$.

- A set is closed if its complement is open.
- The complement of $\{1\}$ is $Z \setminus \{1\} = \{2, 3\}$.
- Is the set $\{2, 3\}$ in our list of open sets τ ? No.

- **Conclusion:** Since its complement is not open, the set $\{1\}$ is not a closed set. This is counter-intuitive.

Problem 2: Limits of Sequences Are Not Unique

This is a much more serious problem. Let's consider a very simple sequence: the constant sequence $x_n = 2$ for all n . Intuitively, this sequence should converge to exactly one limit: the point 2.

Let's check the definition. A point p is a limit of a sequence (x_n) if for **every open neighborhood U of p** , the sequence is **eventually in U** . (Meaning after some index N , all x_n for $n > N$ are in U .)

Is 2 a limit? Let's check the open sets containing 2. They are $\{1, 2\}$ and $\{1, 2, 3\}$. Our sequence is always $x_n = 2$. So every term of the sequence is in $\{1, 2\}$ and in $\{1, 2, 3\}$. The condition is satisfied. So, **2 is a limit**.

Is 3 a limit? Let's check the open sets containing 3. There is only one: the entire space $\{1, 2, 3\}$. Is our sequence $x_n = 2$ eventually in $\{1, 2, 3\}$? Yes, every term is in there from the beginning. The condition is satisfied. So, **3 is also a limit!**

We have a constant sequence that converges to two different points. This violates one of the most fundamental properties we expect from the concept of a limit.

What's the root cause? The open sets in this topology lack "resolution." The only open set that contains 3 is the whole space, which is too big to distinguish 3 from anything else. We have no way to find an open set around 3 that "separates" it from the point 2. This is the problem the Hausdorff property fixes.

Example 1.1 (Verifying the Topology on Z). Let's verify that $\tau = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}\}$ is actually a topology on $Z = \{1, 2, 3\}$.

Step 1: Check $\emptyset$ and Z are in τ . $\emptyset \in \tau$ by definition, and $\{1, 2, 3\} = Z \in \tau$. ✓

Step 2: Check arbitrary unions. List all possible unions:

- $\emptyset \cup \text{anything} = \text{anything} \in \tau$
- $\{1\} \cup \{1, 2\} = \{1, 2\} \in \tau$
- $\{1\} \cup \{1, 2, 3\} = \{1, 2, 3\} \in \tau$
- $\{1, 2\} \cup \{1, 2, 3\} = \{1, 2, 3\} \in \tau$

All unions remain in τ . ✓

Step 3: Check finite intersections.

- $\{1\} \cap \{1, 2\} = \{1\} \in \tau$
- $\{1\} \cap \{1, 2, 3\} = \{1\} \in \tau$
- $\{1, 2\} \cap \{1, 2, 3\} = \{1, 2\} \in \tau$

All finite intersections remain in τ . ✓

Conclusion: τ is a valid topology.

Example 1.2 (Another Non-Hausdorff Space: The Sierpiński Space). Let $S = \{0, 1\}$ with topology $\tau = \{\emptyset, \{0\}, \{0, 1\}\}$.

Step 1: Verify this is a topology. Similar to above, this satisfies all axioms.

Step 2: Check if Hausdorff. Can we separate 0 and 1?

- Open sets containing 0: $\{0\}, \{0, 1\}$
- Open sets containing 1: $\{0, 1\}$ (only one!)

Any open set containing 1 must also contain 0. There are **no disjoint neighborhoods!**

Step 3: Check limit uniqueness. The constant sequence $x_n = 0$ converges to:

- 0? Yes, it's always in $\{0\}$.
- 1? Yes, it's always in $\{0, 1\}$.

Again, non-unique limits!

Conclusion: The Sierpiński space is not Hausdorff.

⚠ Common Mistake

Common Mistake: Assuming all topologies are Hausdorff. Many "natural" examples (like metric spaces) are Hausdorff, which can create false intuition. Always check the definition! The cofinite topology on an infinite set (see Practice Problem 1) is another important non-Hausdorff example.

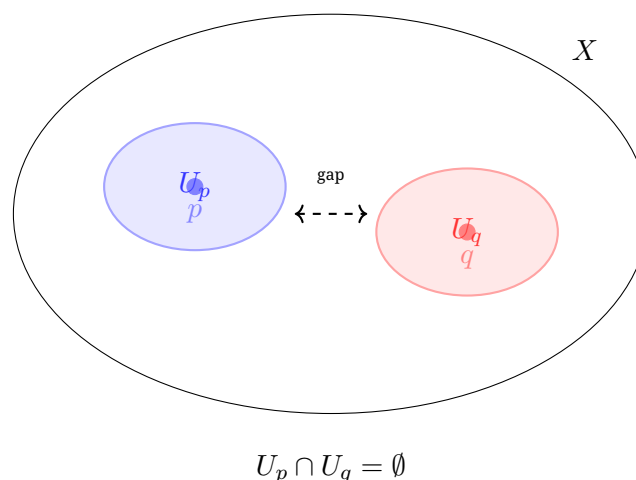
2 The Hausdorff Property (or T_2)

🧠 Intuition

A Hausdorff space is one where any two distinct points can be "housed off" from each other — each gets its own private neighborhood that doesn't intersect with the other's. This is the minimum separation we need for limits to behave sensibly.

📖 Definition: Hausdorff Space (T_2)

A topological space X is called a **Hausdorff space** if for any two distinct points $p, q \in X$, there exist **disjoint open neighborhoods** U_p and U_q of p and q respectively. That is, $p \in U_p$, $q \in U_q$, and $U_p \cap U_q = \emptyset$.



The simple idea is that a space is Hausdorff if it has enough "zoom" or "resolution" to place any two points in their own separate, non-overlapping open "bubbles."

Example 2.1 ($\mathbb{R}$ is Hausdorff — Concrete Verification). Let $p = 2$ and $q = 5$ in $\mathbb{R}$. Find disjoint open neighborhoods.

Step 1: Calculate the distance. $d(p, q) = |5 - 2| = 3$

Step 2: Choose radii. Let $\epsilon = \delta = 1$ (anything less than $3/2 = 1.5$ works).

Step 3: Define neighborhoods.

- $U_p = (2 - 1, 2 + 1) = (1, 3)$

- $U_q = (5 - 1, 5 + 1) = (4, 6)$

Step 4: Verify disjointness. $U_p \cap U_q = (1, 3) \cap (4, 6) = \emptyset$. ✓

Conclusion: We found disjoint open neighborhoods. This works for any distinct points in $\mathbb{R}$.

Example 2.2 (The Plane $\mathbb{R}^2$ is Hausdorff). Let $p = (0, 0)$ and $q = (3, 4)$ in $\mathbb{R}^2$.

Step 1: Calculate Euclidean distance. $d(p, q) = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = \sqrt{25} = 5$

Step 2: Choose radii. Let $\epsilon = \delta = 2$ (less than $5/2 = 2.5$).

Step 3: Define open balls.

- $U_p = B((0, 0), 2) = \{(x, y) : x^2 + y^2 < 4\}$

- $U_q = B((3, 4), 2) = \{(x, y) : (x - 3)^2 + (y - 4)^2 < 4\}$

Step 4: Verify disjointness. The farthest point in U_p from origin has distance < 2 . The closest point in U_q to origin: center is at distance 5, so closest point has distance $> 5 - 2 = 3 > 2$. No overlap! ✓

Conclusion: $\mathbb{R}^2$ is Hausdorff.

Section 2 Summary: The Hausdorff Property

Key Definition: A space X is **Hausdorff** (T_2) if for any $p \neq q$, there exist disjoint open neighborhoods $U_p \ni p$ and $U_q \ni q$.

Key Examples:

- $\mathbb{R}^n$ with standard topology: Hausdorff □
- Discrete spaces: Hausdorff □
- Sierpiński space $\{0, 1\}$ with $\tau = \{\emptyset, \{0\}, \{0, 1\}\}$: Not Hausdorff □
- 3-point space $Z = \{1, 2, 3\}$ with $\tau = \{\emptyset, \{1\}, \{1, 2\}, Z\}$: Not Hausdorff □

Visual Criterion: Can you draw non-overlapping "bubbles" around any two distinct points?

3 Examples of Hausdorff Spaces

Intuition

Metric spaces are the "gold standard" for nice topological spaces. The distance function gives us a precise way to measure "closeness" and construct separating neighborhoods. It turns out that having a metric automatically makes you Hausdorff!

Metric Spaces are Hausdorff

Every metric space (with the topology induced by its metric) is a Hausdorff space. This includes all Euclidean spaces $\mathbb{R}^n$.

- **The Idea:** If two points are different, they have some distance between them. We can just draw small-enough open balls around them so they don't overlap.
- **Proof Sketch:** Let (M, d) be a metric space, and pick two distinct points $p, q \in M$.
- Because $p \neq q$, their distance $d(p, q) = r$ must be strictly positive ($r > 0$).
- Let's choose a radius $\epsilon = r/3$.
- Consider the open ball $U_p = B(p, \epsilon)$ and the open ball $U_q = B(q, \epsilon)$. These are our open neighborhoods.
- We can prove by contradiction using the triangle inequality that these balls are disjoint. If there were a point z in their intersection, then $d(p, z) < r/3$ and $d(q, z) < r/3$. The triangle inequality states $d(p, q) \leq d(p, z) + d(z, q)$, which would mean $r < r/3 + r/3 = 2r/3$. This is false.
- Therefore, $U_p \cap U_q = \emptyset$, and the space is Hausdorff.

Important Consequence: If a space is not Hausdorff, it cannot be a metric space (it is not "metrizable"). Our weird 3-point space Z is an example.

Detailed Proof: Metric Spaces are Hausdorff. **Step 1: Setup.** Let (M, d) be a metric space and let $p, q \in M$ with $p \neq q$.

Step 2: Use the metric. Since $p \neq q$, we have $d(p, q) > 0$. Let $r = d(p, q)$.

Step 3: Construct neighborhoods. Choose $\epsilon = r/3 > 0$. Define:

$$U_p = B(p, \epsilon) = \{x \in M : d(x, p) < \epsilon\}$$

$$U_q = B(q, \epsilon) = \{x \in M : d(x, q) < \epsilon\}$$

Step 4: Show disjointness by contradiction. Suppose $z \in U_p \cap U_q$. Then:

- $d(z, p) < \epsilon = r/3$
- $d(z, q) < \epsilon = r/3$

By the triangle inequality:

$$r = d(p, q) \leq d(p, z) + d(z, q) < \frac{r}{3} + \frac{r}{3} = \frac{2r}{3}$$

This gives $r < 2r/3$, a contradiction since $r > 0$.

Step 5: Conclusion. $U_p \cap U_q = \emptyset$. Thus (M, d) is Hausdorff. □

Example 3.1 (Concrete Metric Space Example: $\mathbb{R}^2$ with Taxi Cab Metric). Let $d_1((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + |y_1 - y_2|$ be the taxicab metric on $\mathbb{R}^2$.

Step 1: Choose distinct points. Let $p = (0, 0)$ and $q = (4, 0)$.

Step 2: Calculate distance. $d_1(p, q) = |4 - 0| + |0 - 0| = 4$.

Step 3: Construct balls. With $\epsilon = 1$:

- $U_p = \{(x, y) : |x| + |y| < 1\}$ (diamond shape)
- $U_q = \{(x, y) : |x - 4| + |y| < 1\}$ (diamond centered at $(4, 0)$)

Step 4: Verify disjointness. The rightmost point of U_p satisfies $|x| + |y| = 1$ with $x > 0$, so $x \leq 1$. The leftmost point of U_q satisfies $|x - 4| + |y| = 1$ with $x < 4$, so $4 - x \leq 1$, meaning $x \geq 3$. Since $1 < 3$, the diamonds don't touch! ✓

Conclusion: Even with a different metric, we get the Hausdorff property.

✍ Discrete Spaces are Hausdorff

Recall that in a discrete space, *every* subset is an open set.

- Let p, q be two distinct points in a discrete space X .
- We need to find disjoint open neighborhoods. This is easy!
- The set $U_p = \{p\}$ is an open set, since every set is open.
- The set $U_q = \{q\}$ is also an open set.
- Since $p \neq q$, their intersection is $\{p\} \cap \{q\} = \emptyset$.
- Therefore, every discrete space is Hausdorff.

Example 3.2 (Finite Discrete Space). Let $X = \{a, b, c\}$ with the discrete topology $\tau = \mathcal{P}(X)$ (all subsets).

Step 1: Check separation of a and b .

- $U_a = \{a\}$ is open
- $U_b = \{b\}$ is open
- $U_a \cap U_b = \emptyset$ ✓

Step 2: Similarly for other pairs.

- a and c : use $\{a\}$ and $\{c\}$
- b and c : use $\{b\}$ and $\{c\}$

Conclusion: Every pair of distinct points has disjoint open neighborhoods. X is Hausdorff.

📋 Section 3 Summary: Examples of Hausdorff Spaces

Key Theorem: Every metric space is Hausdorff.

Proof Strategy: Use the metric to find the distance $r = d(p, q) > 0$, then take balls of radius $r/3$. Triangle inequality ensures they're disjoint.

Key Examples:

- $\mathbb{R}^n$ with Euclidean metric: Hausdorff □
- $\mathbb{R}^n$ with taxicab metric: Hausdorff □
- Any discrete space: Hausdorff □ (use singletons)
- Any indiscrete space with $|X| \geq 2$: Not Hausdorff □

Key Consequence: Non-Hausdorff spaces cannot be metrizable.

4 The Payoff: "Nice" Properties of Hausdorff Spaces

Intuition

Now we reap the benefits! The Hausdorff condition fixes the two weird behaviors we saw in our 3-point space: (1) points might not be closed sets, and (2) sequences might converge to multiple limits. In Hausdorff spaces, both problems vanish.

Proposition 1: In a Hausdorff space

This implies that every singleton set $\{p\}$ is a closed set, fixing the first "weird" property we saw.

Proof Idea: We want to show a singleton $\{p\}$ is closed. The easiest way is to show its complement, $X \setminus \{p\}$, is **open**. To show a set is open, we can show that every point inside it has an open neighborhood that is also contained within the set.

Walkthrough:

1. Let X be a Hausdorff space and let $p \in X$. We want to prove $X \setminus \{p\}$ is open.
2. Pick any arbitrary point $q \in X \setminus \{p\}$. This just means $q \neq p$.
3. By the Hausdorff property, there exist disjoint open sets U_q and U_p such that $q \in U_q$ and $p \in U_p$.
4. Since U_q and U_p are disjoint, the point p cannot be in U_q . This means that the entire open neighborhood U_q is contained within the complement of $\{p\}$. That is, $U_q \subseteq X \setminus \{p\}$.
5. We have shown that for *every* point q in the complement, we can find an open bubble around it that is still inside the complement.
6. Therefore, the complement $X \setminus \{p\}$ can be written as the union of all these open bubbles ($\bigcup_{q \neq p} U_q$), and an arbitrary union of open sets is always open.
7. Thus, $X \setminus \{p\}$ is open, which means $\{p\}$ is closed.
8. A finite set is a finite union of singletons (points). A finite union of closed sets is always closed, so any finite set is also closed.

Detailed Proof: Finite Sets are Closed in Hausdorff Spaces. **Step 1: Show singletons are closed.** Let $p \in X$ where X is Hausdorff. We'll show $\{p\}$ is closed by showing $X \setminus \{p\}$ is open.

Step 2: Take arbitrary $q \neq p$. For each such q , by Hausdorff property, $\exists$ disjoint open sets $U_q \ni q$ and $V_q \ni p$.

Step 3: Key observation. Since $U_q \cap V_q = \emptyset$ and $p \in V_q$, we have $p \notin U_q$. Therefore $U_q \subseteq X \setminus \{p\}$.

Step 4: Express complement as union.

$$X \setminus \{p\} = \bigcup_{q \neq p} U_q$$

Step 5: Conclude open. Each U_q is open, and arbitrary unions of open sets are open. Thus $X \setminus \{p\}$ is open, so $\{p\}$ is closed.

Step 6: Extend to finite sets. Let $F = \{p_1, \dots, p_n\}$ be finite. Then:

$$F = \bigcup_{i=1}^n \{p_i\}$$

is a finite union of closed sets, hence closed. □ □

Example 4.1 (Verifying: Points are Closed in $\mathbb{R}$). In $\mathbb{R}$, $\{0\}$ should be closed.

Step 1: Check complement. $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$.

Step 2: Is complement open? Yes! $(-\infty, 0)$ and $(0, \infty)$ are both open intervals.

Step 3: Conclusion. $\{0\}$ is closed. This matches our proposition since $\mathbb{R}$ is Hausdorff.

💡 Proposition 2: In a Hausdorff space

This fixes the most serious "weird" property we found.

Proof by Contradiction:

1. Let (x_n) be a sequence in a Hausdorff space X . Suppose, for the sake of contradiction, that the sequence converges to two distinct limits, p and q , where $p \neq q$.
2. Because X is Hausdorff, we can find disjoint open neighborhoods U_p containing p and U_q containing q , so $U_p \cap U_q = \emptyset$.
3. Since $x_n \rightarrow p$, the sequence must eventually be in U_p . This means there is some integer N_p such that for all $n > N_p$, $x_n \in U_p$.
4. Similarly, since $x_n \rightarrow q$, there is some integer N_q such that for all $n > N_q$, $x_n \in U_q$.
5. Now, pick an integer N that is larger than both N_p and N_q (let $N = \max(N_p, N_q)$).
6. For any $n > N$, the term x_n must be in U_p (from step 3) **and** it must be in U_q (from step 4).
7. This means $x_n \in U_p \cap U_q$.
8. But this is a **contradiction!** We chose U_p and U_q to be disjoint, so their intersection is empty.
9. Our initial assumption that there were two distinct limits must be false. Therefore, the limit is unique.

Detailed Proof: Unique Limits in Hausdorff Spaces. **Step 1: Setup for contradiction.** Let (x_n) be a sequence in Hausdorff space X . Suppose $x_n \rightarrow p$ and $x_n \rightarrow q$ with $p \neq q$.

Step 2: Apply Hausdorff property. Since $p \neq q$, there exist disjoint open sets $U_p \ni p$ and $U_q \ni q$ with $U_p \cap U_q = \emptyset$.

Step 3: Use convergence to p . Since $x_n \rightarrow p$, by definition:

$$\exists N_p \in \mathbb{N} \text{ such that } \forall n > N_p, x_n \in U_p$$

Step 4: Use convergence to q . Since $x_n \rightarrow q$, by definition:

$$\exists N_q \in \mathbb{N} \text{ such that } \forall n > N_q, x_n \in U_q$$

Step 5: Find the contradiction. Let $N = \max(N_p, N_q)$. Then for all $n > N$:

$$x_n \in U_p \quad \text{and} \quad x_n \in U_q$$

Thus $x_n \in U_p \cap U_q = \emptyset$. Contradiction!

Step 6: Conclusion. Our assumption $p \neq q$ is false. Thus $p = q$ and limits are unique. □

Example 4.2 (Unique Limits in $\mathbb{R}$). Consider the sequence $x_n = \frac{1}{n}$ in $\mathbb{R}$.

Step 1: Check convergence. We know $x_n \rightarrow 0$.

Step 2: Could it converge to another point? Suppose $x_n \rightarrow L$ where $L \neq 0$.

Step 3: Apply Hausdorff separation. Since $\mathbb{R}$ is Hausdorff, we can find disjoint neighborhoods $U_0 \ni 0$ and $U_L \ni L$.

Step 4: Derive contradiction. Since $x_n \rightarrow 0$, eventually all $x_n \in U_0$. Since $x_n \rightarrow L$, eventually all $x_n \in U_L$. But $U_0 \cap U_L = \emptyset$! Impossible.

Conclusion: The only possible limit is 0.

Common Mistake

Common Mistake: Assuming that in a Hausdorff space, **all** sequences converge. This is false! The Hausdorff property guarantees **uniqueness** of limits (if they exist), not **existence**. For example, in $\mathbb{R}$, the sequence $x_n = n$ does not converge at all.

Another mistake: Forgetting that the Hausdorff property is about **distinct** points. You cannot separate a point from itself!

Section 4 Summary: Nice Properties of Hausdorff Spaces

Proposition 1: In a Hausdorff space, all finite sets are closed.

Proof Strategy: Show complement is open by covering it with neighborhoods that avoid the point.

Proposition 2: In a Hausdorff space, convergent sequences have unique limits.

Proof Strategy: Assume two distinct limits, get disjoint neighborhoods, derive contradiction via eventual membership in both.

Key Contrast with Non-Hausdorff:

Property	Hausdorff	Non-Hausdorff (e.g., $\mathbb{Z}$)
Points closed?	Yes	Not necessarily
Unique limits?	Yes	No
Metrizable?	Possible	Impossible

Practice Problems

- Cofinite Topology:** Let X be an infinite set. The cofinite topology on X is defined by declaring a set $U \subseteq X$ to be open if it is the empty set or if its complement $X \setminus U$ is finite. Is X with the cofinite topology a Hausdorff space? Prove your answer. (Hint: Can you find disjoint open sets around two points?)
- Subspaces:** Let X be a Hausdorff space and let $A \subseteq X$ be a subset. Show that A with the subspace topology is also a Hausdorff space.

3. Let $X = \{a, b, c\}$. Determine if the space is Hausdorff for each of the following topologies:
- a) $\tau_1 = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{a, b, c\}\}$
 - b) $\tau_2 = \{\emptyset, \{a\}, \{b, c\}, \{a, b, c\}\}$
 - c) The discrete topology.

Further Reading

- **James Munkres, *Topology* (2nd Edition):** See Chapter 2, Section 17 for a formal treatment of Hausdorff spaces and the other separation axioms.